

Foreword

These course notes represent my efforts to edit, summarize, and catalog the lectures for the Software Architecture and Design (CS-6310) course at Georgia Tech, taken in Spring 2026 as part of Tech's innovative, fully online master of science in computer science (OMSCS) program. The original lectures are [freely available online](#) and presented by Dr. Spencer Rugaber, as open-source content.

There are over 35 quizzes that comprise 5% of the grade based on the lectures, designated as a form of “padding” the grade according to the TAs. There is a group project with group members organically organized through forum posts, usually within the same time zone to facilitate availability for online meeting times. These groups and their projects tend to have high variability in quality and effect, such as determining leadership, which skills each team member possesses, how they can interact and allocate time over the semester while most (80%) of the students are working in industry, etc. There are also two open-ended exams, with the second exam optional depending on the student's performance on the first exam. Given these course requirements, and the academic rigor of a “superelite” Institute of Technology like Georgia Tech, the barriers to learn this course's material in a formal, academic setting are relatively high. However, the course material, especially the lectures, are valuable enough to others that a condensed, edited, visual reference and guide would be a godsend. Effective and efficient software systems form an area of critical need in our society today, and software architecture training provides a tried-and-true method of enabling the development and execution of these large-scale software systems. These systems can span millions of lines of code, tens or hundreds of modules, and model and run large, essential infrastructure projects like air traffic control and routing or power plant operations.

In industry, software system architect is generally understood as the role superior to that of senior software developer, regardless of title. The software system architect works with management and customers to define technical specifications and high-level software solutions, while also directing a team's or teams' development throughout the project from a technical standpoint. I took this class to get a head start on the next level of work that would be available to me after working as a senior developer for some time; it is difficult to get training or experience in industry on software architecture topics unless the company promotes or authorizes an individual, while Georgia Tech provides this class to anyone enrolled in the OMSCS program – it is never full or impacted, with no prerequisites beyond admission. These notes could also provide awareness and perspective to those working at technology firms in the industry who do not work with code directly, to give insight into how these applications are made from a high-level perspective.